

Numerical Methods

BCA — Semester IV — 2022 — Answer Sheet
Subject Code: CACS252

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group C Attempt any TWO questions[2x10=20]

2. If the true value of π is 3.1415926 and its approximate value is given by 3.1428571. Find the absolute and relative errors.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What are the advantages of Lagrange's formula over Newton's formula? When Newton's forward interpolation formula is used?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Find the polynomial that interpolates $f(0) = 0$, $f(1) = 1$, $f(2) = 0$, $f(3) = 1$, $f(4) = 0$ using Newton divided differences. Use the Newton table to generate the necessary divided differences.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Using the trapezium rule, evaluate the integral $I = \int_0^1 \frac{dx}{x^2 + 6x + 10}$ with 2 and 4 sub intervals. Compare with the exact solution. Comment on the magnitudes of the errors obtained.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. For solving a linear system, compare Gauss elimination method and Gauss Jordan method. Why Gauss Seidel method is better than Jacobi's iterative method?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Using Euler's method find $y(0.2)$ from $\frac{dy}{dx} = x + y$, $y(0) = 1$ with $h = 0.1$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Solve the Poisson equation $\Delta^2 f = 2x^2 y^2$ over the square domain $0 \leq x \leq 3$ and $0 \leq y \leq 3$ with $f = 0$ on the boundary and $h = 1$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. When would we not use N-R method? Find the root of the equation $x^3 - 4x + 1 = 0$, lying in $(0, 1)$ using Bisection method performing 10 iterations.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Find the dominant eigenvalue of $A = \begin{bmatrix} 2 & 3 \\ 5 & 4 \end{bmatrix}$ by power method. Apply Gauss Seidel method to solve system of equations: $6x_1 - 2x_2 + x_3 = 11$; $-2x_1 + 7x_2 + 2x_3 = 5$; $x_1 + 2x_2 - 5x_3 = -1$, with the initial vector of $(0, 0, 0)$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Prepare the multi-step methods available for solving ordinary differential equations. Evaluate the value of y at $x = 0.1$ and 0.2 to 4 decimal places given $\frac{dy}{dx} = x^2 y - 1$ $\quad y(0) = 1$, using Taylor series method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.